METHOD VALIDATION OF SD BIOSENSOR'S STANDARD™ F PLATFORM FOR THE RAPID DETECTION AND DIFFERENTIATION OF DENGUE NS1 ANTIGEN, DENGUE IgM/IgG, ZIKA ANTIGEN AND ZIKA IgM/IgG ANTIBODIES IN HUMAN PLASMA AND WHOLE BLOOD SAMPLES

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1. Introduction & Research Problem

- Dengue virus (DENV) and Zika virus (ZIKV) are co-circulating flaviviruses transmitted by *Aedes* mosquitoes.
- They share non-specific early clinical presentations such as fever, rash, and myalgia, making accurate symptom-based diagnosis difficult.
- Severe outcomes like Dengue Hemorrhagic Fever or Zika-associated congenital microcephaly necessitate rapid, reliable differential diagnosis.
- **Research Problem:**
  - Reference diagnostics like RT-PCR and ELISAs offer high accuracy but are resource- and labour-intensive, requiring specialized infrastructure and long runtimes.
  - Furthermore, antibody cross-reactivity among flaviviruses complicates traditional serological interpretations[1].

2. Research Question & Objective

Research Question:

- Can the rapid SD Biosensor STANDARD™ F fluorescent immunoassay platform offer a more rapid, accurate, and differential diagnosis of arboviral infections (Dengue and Zika) compared to reference gold-standard methods (RT-PCR and ELISA) in a resource-limited laboratory setting?

Objective:

- To validate the diagnostic accuracy, positive percent agreement (PPA), and negative percent agreement (NPA) of the STANDARD™ F platform against reference ELISA and RT-PCR methods using 595 archived patient samples in a resource-limited referral setting in Jamaica.

3. Project Pipeline

Sample Retrieval & Blinding
↓
Reagent Preparation & Loading
↓
Incubation & Lateral Flow
↓
Automated Optical Analysis[2]
↓
Statistical Performance Evaluation





4. Dataset & Results

Cohorts & Demographics:

- Analyzed **595 archived patient samples** (245 DENV, 350 ZIKV; 82.12% female, median age 23.88 years).

Baseline Positivity (Reference Assays):

- **DENV ELISA:** 48.97% positive (n=120).
- **ZIKV ELISA:** 28.02% positive (n=44).
- **ZIKV RT-PCR:** 47.66% positive (n=92).

SDB-SF FIA test	PPA/Sensitivity % (95% CI)	NPA/Specificity % (95% CI)	Overall % (95% CI)	Reference/Gold Standard
DENV IgG	83.33 (51.59 - 97.91)	100 (2.50 - 100)	84.61 (54.55 - 98.08)	Focus™ Dengue IgM (ELISA) Sens = 96%; Spec= 97%, SD Bioline™ Dengue IgM/IgG (ELISA) Sens = 94.6%; Spec = 96.5%.
DENV NS1	63.63 (53.92 - 72.60)	99.19 (95.59 - 99.98)	82.47 (76.99 - 87.12)	Roche™ Lightmix® Dengue (PCR) 100% for both Sens & Spec.
ZIKV IgM	2.63 (0.07 - 13.81)	100 (96.73 - 100)	75.17 (67.43 - 81.87)	Euroimmun™ Anti-Zika Sens = N/A; Spec = 98-100%.
ZIKV IgG	0 (0 - 45.93)	100 (15.81 - 100)	25 (3.19 - 65.09)	Virus IgM/IgG (ELISA).
ZIKA Antigen	2.17 (0.26 - 7.63)	99.01 (94.61 - 99.97)	52.85 (45.55 - 60.06)	Lanciotti Singleplex™ (PCR); 100% for both Sens & Spec.

Table 1. Comparison of the diagnostic performance of the individual assay parameters in comparison to the reference/gold standard methods. CI = confidence interval; Sens = Sensitivity; Spec = Specificity 95% Confidence Interval is shown in parenthesis. The SD Biosensor STANDARD™ F platform exhibited excellent specificity (99-100%) across all parameters, reliably ruling out non-cases. However, while Dengue assays demonstrated moderate accuracy (>82%), all Zika parameters failed severely in sensitivity (0-2.63%), highlighting major performance limitations compared to reference RT-PCR and ELISA gold standards.

Operational & Performance Metrics:

- **Invalid Rate =** 0.78% (3/387 invalid tests) due to thick whole-blood sample viscosity interfering with lateral flow migration.
- **Turnaround Time =** 15–20 minutes total run time for STANDARD™ F vs. 75–100 minutes for reference ELISAs and > 60 minutes for RT-PCR workflows.

Co-infection & Dual-Positivity Performance:

- **44 samples** tested positive for both DENV and ZIKV via reference gold standards.
- Upon retesting **44 dual-positive samples** on the STANDARD™ F platform, 0/22 were detected for ZIKV and only 3/22 were detected for DENV, highlighting significant cross-reactivity and sensitivity limitations during co-infections.

5. Biological Interpretation, Conclusion & Future Work

- Moderate DENV NS1 (63.63%) and IgG (83.33%) sensitivity supports rapid endemic triage, though NS1 wanes post-acute phase due to immune complexing.
- Near-zero ZIKV sensitivity (0%–2.63%) reflects low analytical limits of detection or suboptimal antigen binding, making the platform **unreliable**.
- Despite rapid turnaround (15 mins), the platform **failed to match manufacturer claims** (97-100%) or reference gold standards.
- Therefore this platform is **unsuitable** as a standalone replacement for reference arbovirus diagnostics in Jamaica without major manufacturer optimization.
- **Future work** could involve optimizing ZIKV antigen design, adjusting cut-off algorithms, and evaluating combined NS1 + IgM/IgG panels.

6. References

[1](Charrel et al., 2016); [2](SD Biosensor Inc., 2018)

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